

Bioinformatics Bootcamp Assessment 01

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Question 01

A researcher has discovered a DNA sequence that may be associated with a rare genetic disorder. Before conducting laboratory experiments, they want to determine whether this sequence has been reported previously. Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides and how it would assist the researcher.

The researcher should use the following biological databases:

GenBank

GenBank is an open database of DNA and RNA sequences of many organisms.

The researcher may search up the DNA sequence to see if it has been reported before and find related information.

NCBI

NCBI provides access to many biological databases, including GenBank.

It gives researchers one place to look up DNA sequences, genes, scientific articles and information about genetic diseases.

BLAST

BLAST compares the new DNA sequence to sequences already stored in the databases.

It shows if the sequence is same or similar to sequences previously reported and identifies related genes or organisms.

Suppose a researcher finds a new DNA sequence. First, they check GenBank to see if the sequence is already there. Then they compare it with known sequences using BLAST. If they find a similar sequence, they use NCBI to learn more about the gene and any disease associated with it.

Together, these databases help identify the sequence, study its possible function, and investigate its association with the rare genetic disorder.

Question 02

A scientist is studying a disease-causing bacterium and wants to compare its genome with those of related bacterial species to identify conserved genes. Which biological database(s) would you recommend for this analysis? Explain why these databases are appropriate and what information they provide.

The scientist should use these biological databases:

NCBI

- NCBI provides complete bacterial genome sequences, gene information and related research.
- This helps the scientist to find the genomes of the disease-causing bacterium and related species of bacteria.

Genbank

- GenBank contains DNA sequences from many organisms, including bacteria.
- The scientist can download the genome sequences of bacteria and compare them.

BLAST

- BLAST searches for similarities between DNA or protein sequences and sequences in databases.
- It finds sequences that are similar in different species of bacteria and this helps in identifying conserved genes.

Suppose a scientist studying a disease-causing bacterium. They retrieve the genome from GenBank and compare it to the genomes of related bacteria with BLAST. BLAST shows that there are some genes common to all the bacteria. These are conserved genes, which could be important for the bacteria's survival.

The scientist uses these databases to compare bacterial genomes , to identify conserved genes and to better understand the disease-causing bacterium .

Question 03

A biotechnology company has isolated a protein from a newly identified microorganism and wants to investigate whether similar proteins exist in other species. Which biological database(s) would you recommend, and what information would you expect to obtain from each database? Explain your reasoning.

The biotechnology company should utilize these biological databases:

UniProt

UniProt is a protein sequence and functional information database.

It helps the company find other species that have similar proteins and learn what those proteins do.

BLAST

BLAST compares the new protein sequence with proteins already stored in databases.

It tells us that other organisms have similar proteins.

NCBI

NCBI provides Information about proteins, genes, genomes, and scientific studies.

It can help the company learn more about similar proteins and the organisms they come from.

Now imagine the company finds a new protein in a bacterium. They compare its sequence to known proteins using BLAST. BLAST finds a similar protein in some other bacteria . They then check up this protein in UniProt and discover it is involved in the breakdown of toxic chemicals.

The company can use UniProt, BLAST and NCBI to find similar proteins in other species, understand their functions and study their importance.

Question 04

A scientist has identified an unknown protein sequence and wants to determine its function. Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides.

The scientist should to use the following databases:

UniProt

This database holds the sequence of proteins and information of their functions. It will help the scientist understand the function of the protein, localization in the body, and its role in the cell.

BLAST

BLAST compares the unknown protein with other proteins in databases. It will help the scientist find similar proteins whose function is known.

NCBI

This resource holds information on proteins, genes, genomes, and scientific research. It will help the scientist obtain more information about the protein.

The scientist discovers an unknown protein and uses BLAST, finding out that it is highly similar to another protein involved in DNA repair. Then he uses UniProt and finds out that the protein repairs damaged DNA.

With the help of BLAST, UniProt, and NCBI, the scientist can learn about similar proteins and functions of the protein.

Question 05

You have been provided with an unknown biological sequence but are not told whether it is DNA, RNA, or protein.

Describe the step-by-step approach you would follow to identify the sequence using biological databases. Explain which databases you would use at each stage and why.

If I want to identify an unknown biological sequence I would follow these steps:

Step 1: Check the type of sequence

I would begin by looking at the letters in the sequence.

If it has A, T, G, C, it's probably DNA.

If it contains A, U, G, C, it is probably RNA .

If it contains many different letters (like M, L, K, F, R) it is most likely a protein.

Step 2: Search the sequence using BLAST

I would use BLAST to compare the unknown sequence with sequences in biological databases.

BLAST helps in identification of the sequence and finds similar sequences of DNA, RNA or proteins .

Step 3: Check NCBI

NCBI provides information about genes, proteins, genomes, and scientific research.

Helps to identify the organism the sequence is from , and gives more details .

Step 4: Use the right database

If the sequence is DNA or RNA, I would use GenBank to look up nucleotide sequences that match.

If it is a protein, I would go to UniProt and find its function and associated proteins.

Suppose that I have an unknown sequence. I search it with BLAST and discover that it is a bacterial protein. Then I look in UniProt and learn that the protein plays a role in resistance to antibiotics.

By using **BLAST**, **NCBI**, **GenBank**, and **UniProt**, I can identify whether the sequence is DNA, RNA, or protein, determine which organism it comes from, and understand its function.